

Research Readiness Statement

FINESST companion document (1-page limit + graduate study timeline). This section is NOT anonymized — it names the FI and institution. Items in [brackets] are placeholders the FI must fill in personally; do not submit with brackets remaining.

Preparation

Degree program. [FI name] is a [PhD/MS] student in [department], [university], enrolled since [term/year], expected graduation [term/year]. [Advisor name] ([title]) is the proposal PI.

Research experience. The FI independently designed and built an end-to-end terrain-analysis pipeline on airborne lidar for an unrelated landscape (detecting historical disturbance features — excavated pits, access roads, and cleared pads — across the Appalachian Plateau): raw point-cloud processing (PDAL), bare-earth derivative generation (GDAL/WhiteboxTools), hand-labeled training-set construction in QGIS, U-Net and instance-segmentation model training and honest per-instance evaluation (PyTorch), and DEM-of-Difference change analysis with ICP co-registration and robust (NMAD-based) uncertainty. This is the same method stack the proposed project uses: every tool named in the S/T/M methodology is one the FI already operates end to end.

Coursework completed. [List: e.g., remote sensing, GIS, statistics/ML courses with numbers and grades — pull from transcript.]

Coursework planned. [Bayesian / hierarchical statistical modeling course, term]; [polar/cold-regions geomorphology or hydrology course or directed reading with the PI, term]. These directly close the two stated skill gaps: hierarchical driver modeling (O1) and Antarctic process knowledge.

Skills inventory. Python scientific stack, PDAL/GDAL/WhiteboxTools, PyTorch, QGIS; version-controlled, documentation-first research practice (public repository, per-experiment write-ups, append-only analysis log). [Add: any field experience, presentations, publications, teaching.]

Why ready now. The proposed work begins with data acquisition and pipeline construction from public archives — the exact activity the FI's prior project consisted of. The new elements (Antarctic hydrology context, hierarchical modeling) are scheduled as Year-1 coursework and PI mentorship, before the analyses that need them.

Graduate study timeline

Item	Detail
Degree sought	[PhD/MS], [subject area]
Enrolled since	[term, year]
Qualifying/candidacy exam	[completed date / planned term]
Proposed FINESST period	Award start + 3 years
Expected graduation	[term, year — must accommodate the 3-year award]
Milestones during award	Yr 1: coursework + attribution manuscript; Yr 2: generalization paper + AGU; Yr 3: synthesis + defense